

ESG 633 Programming for Geoinformatics

Lab 5: Vector Data Processing

There are 2 shape files given with these shape files do the following:

- Meghalaya State
 - NE_states
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1. Print the projection of all shapefiles.
 2. Check whether the shapefiles have missing attributes.
 3. Reproject the CRS of all shapefiles to EPSG:4326, plot it using matplotlib (highlight the states for NE_states shapefile) and save it as a new shapefile.
 4. Calculate the areas of North Eastern India, sort them in the descending order of area and plot a bar chart of North Eastern State areas.
 5. Download your state shape file from “https://gadm.org/download_country.html” and plot it using matplotlib.
 6. Download India shape file from “https://gadm.org/download_country.html” and extract your favorite region shape file and store it in ESG:4326 crs.
 7. Choose 1 state from the shape file and plot the bar plot for visualizing the areas of districts in that state.